CODEFORCES
Sponsored by Telegram

AmPieSamPie | Logout

HOME TOP CATALOG CONTESTS GYM PROBLEMSET GROUPS RATING EDU API CALENDAR HELP

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

C. Fast Sum

time limit per test: 1 second
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

Your friend is trying to solve the following problem. You are given $\mathbf{T}$ test cases. For each test case, you are given an integer $\mathbf{N}$. You have to find out the summation of 1 to N. More formally, your friend has to calcuate

$$\sum_{x=1}^{x=N} x$$

Your friend wrote the following code in Python to solve it:

```python
T = int(input())

for _ in range(T):
    N = int(input())
    sum = 0
    for i in range(1, N + 1):
        sum += i
    print(sum)
```

Same code in Java:

```java
import java.util.Scanner;

public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int T = sc.nextInt();
        for (int t = 0; t < T; t++) {
            int N = sc.nextInt();
            int sum = 0;
            for (int i = 1; i <= N; i++)
                sum += i;
            System.out.println(sum);
        }
    }
}
```

However, the code is not passing the online judge due to some unknown errors for large values of N.

Since you are currently studying CSE221 and have learned about time complexity, help your friend come up with a more efficient solution.

CSE221 Assignment 01 Fall 2026
Contest is running
9 days
Virtual Participation

→ Submit?

Language: Java 21 64bit
Choose file: Choose File No file chosen

Success!
CLOUDFLARE
Privacy • Help

Submit

Input

The first line contains a single integer T $(1 \leq T \leq 10^4)$ — the number of test cases.

The next T lines each contain a single integer N $(1 \leq N \leq 10^6)$.

Output

For each test case, print a single integer — the summation from 1 to N.

Example

input	Copy
5 2 5 10 12 100	

output	Copy
3 15 55 78 5050	

Codeforces (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Oct/23/2026 22:32:55 UTC+6 (f1).
Desktop version, switch to mobile version.
Privacy Policy | Terms and Conditions

Supported by

